

A Statistical Toolkit for Linguistic Typology

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My life in pictures





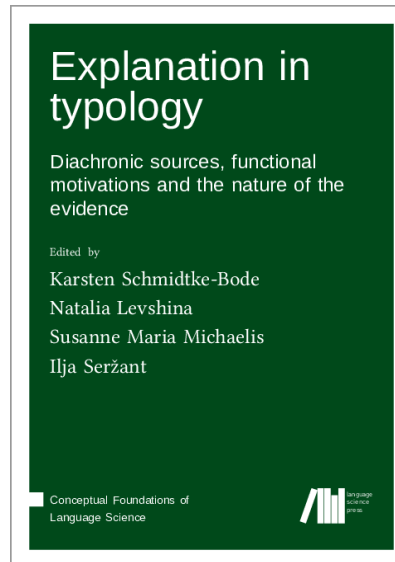
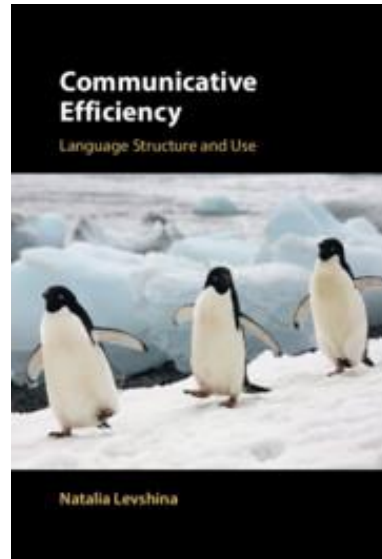
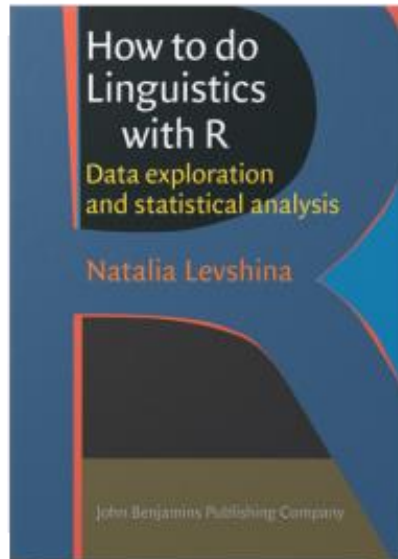
What about you?

Quantitative and corpus-based typology

<https://levshina.github.io/publications/>

DE GRUYTER MOUTON

Linguistic Typology 2019; 23(3): 533–572



Natalia Levshina*

Token-based typology and word order entropy: A study based on Universal Dependencies

DE GRUYTER MOUTON

Linguistic Typology 2021; aop

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Corpus-based typology: applications, challenges and some solutions

Cross-Linguistic Trade-Offs and Causal Relationships Between Cues to Grammatical Subject and Object, and the Problem of Efficiency-Related Explanations

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Published by De Gruyter Mouton

Volume 7 Issue s3 - Efficiency in human languages: corpus evidence for universal principles / edited by Natalia Levshina and Steven Moran

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Issue of *Linguistics Vanguard*



What this course is about

Typologist's Goals	Method
test cross-linguistic generalizations (biases, correlations and implications)	mixed-effects phylogenetic/phylogeographic regression
explore cross-linguistic semantic dimensions of words or constructions, compare form- meaning mappings	probabilistic semantics maps with Multidimensional Scaling
compare formal and functional constraints on formal variation across languages	conditional inference trees and random forests

Plan (tentative)

- Session 1: testing cross-linguistic generalizations
- Session 2: creating probabilistic semantic maps
- Session 3:
 - comparing constraints
 - discussing your own ideas

Course materials

- See the repository: <https://github.com/levshina/Frejus2026>

Outline

- Cross-linguistic generalizations and problems of testing them
- An overview of different solutions
- A very quick and dirty introduction to Bayesian statistics
- A hands-on case study with brms

Have you heard of any language universals?

Univariate biases

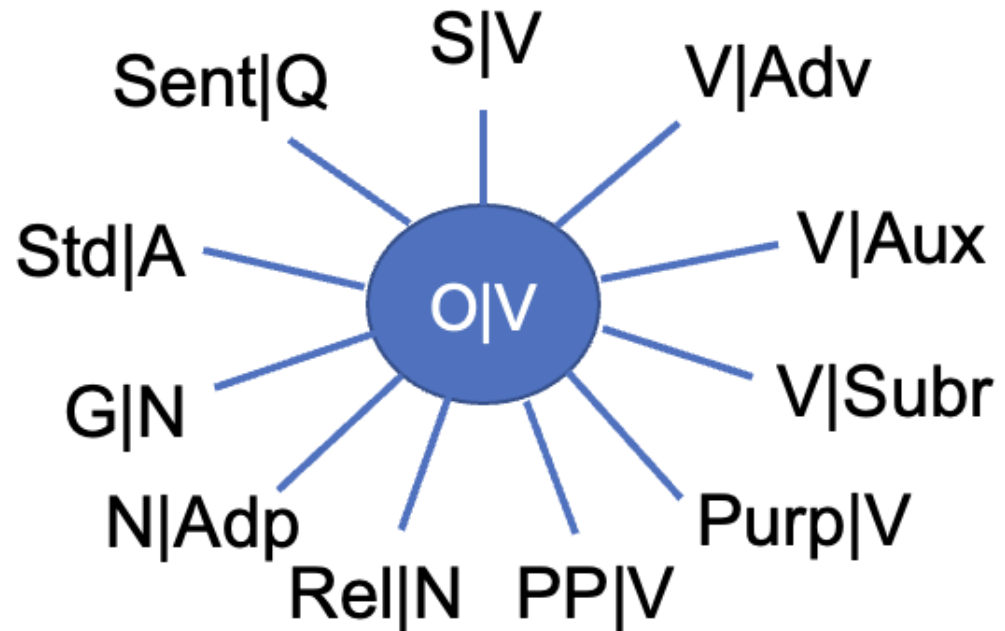
- Greenberg's universal 1: "In declarative sentences with nominal subject and object, the dominant order is almost always one in which the subject precedes the object."

	Subject - Object	Object - Subject
Object - Verb	Yes	No

Testable hypothesis:

A language is more likely to have SO than OS.

Correlations: Many WO universals



Croft 2003, based on Dryer 1992

	Noun - Postposition	Preposition - Noun
Object - Verb	Yes	No
Verb - Object	No	Yes

Testable hypothesis:

If a language has OV, then it has higher chances of having postpositions than if it has VO. (or the other way round)

Implications: Greenberg's Universal 25

"If the pronominal object follows the verb, so does the nominal object."

	Verb - Nom. Object	Nom. Object - Verb
Pron. Object - Verb	Yes	Yes
Verb - Pron. Object	Yes	No

Testable hypothesis

N languages with Verb + PronObj AND Verb + NomObj
is GREATER THAN
N languages with Verb + PronObj AND NomObj + Verb)

How to test?

- Greenberg looked at a sample of 30 languages, no statistical tests.
- Using this method, one can only test exceptionless universals (rare!)

But how to test **statistical** correlations and implications?

Traditional statistical tests

- Popular in computational linguistics

Linguistic Complexity: Relationships Between Phoneme Inventory Size, Syllable Complexity, Word and Clause Length, and Population Size

Gertraud Fenk-Oczlon^{1} and Jürgen Pilz²*

Method

- "The parallel textual material used for our analysis consists of 22 simple declarative sentences encoding one proposition and using basic vocabulary. "
- "Native speakers of 61 languages from 18 language families and from all continents were asked to translate the 22 sentences into their mother tongue. Most of our informants were students, many of them linguists we met at international conferences. The basic requirement was a good knowledge of either English or German in order to be able to translate the test sentences into their mother tongue."

Language sample

Athabaskan-Eyak-Tlingit [Navajo (NAV)] **Atlantic-Congo** [Bafut (BFD), Ewondo (EWO), Lamnso (LNS), Kirundi (RUN), Yoruba (YOR)] **Austronesian** [Batak (BYA), Cham (CJA), Chuukese (CHK), Hawaiian (HAW), Javanese (JAV), Kadazan (DTB), Kemak (KEM), Malagasy (BHR), Malay (MEO), Mambae (MGM), Minangkabau (MIN), Nias (NIA), Roviana (RUG), Tagalog (TGL)] **Austroasiatic** [Vietnamese (VIE)] **Basque** [Basque (EUS)] **Chiquitano** [Chiquitano (CAX)] **Dravidian** [Telugu (TEL)] **Indo-European** [Albanian (SQI), Armenian (XCL), Bulgarian (BUL), Czech (CES), Croatian (HLV), Dutch (NLD), English (ENG), French (FRA), German (DEU), Greek (ELL), Hindi (HIN), Icelandic (ISL), Italian (ITA), Latvian (LAV), Macedonian (MKD), Norwegian (NOR), Panjabi (PAN), Persian (PER), Polish (POL), Portuguese (POR), Romanian (RON), Russian (RUS), Slovenian (SLV), Spanish (SPA), Tajik (TGK)] **Japonic** [Japanese (JPN)] **Kartvelian** [Georgian (GEO)] **Koreanic** [Korean (KOR)], **Mande** [Bambara (BAM)] **Sino-Tibetan** [Mandarin Chinese (CMN)] **Tai-Kadai** [Thai (THA)] **Turkic** [Turkish (TUR)] **Uralic** [Estonian (EKK), Finnish (FIN), Hungarian (HUN)] **Uto-Aztecan** [Hopi (HOP)] **Western Daly** [Maranunggu (ZMR)].

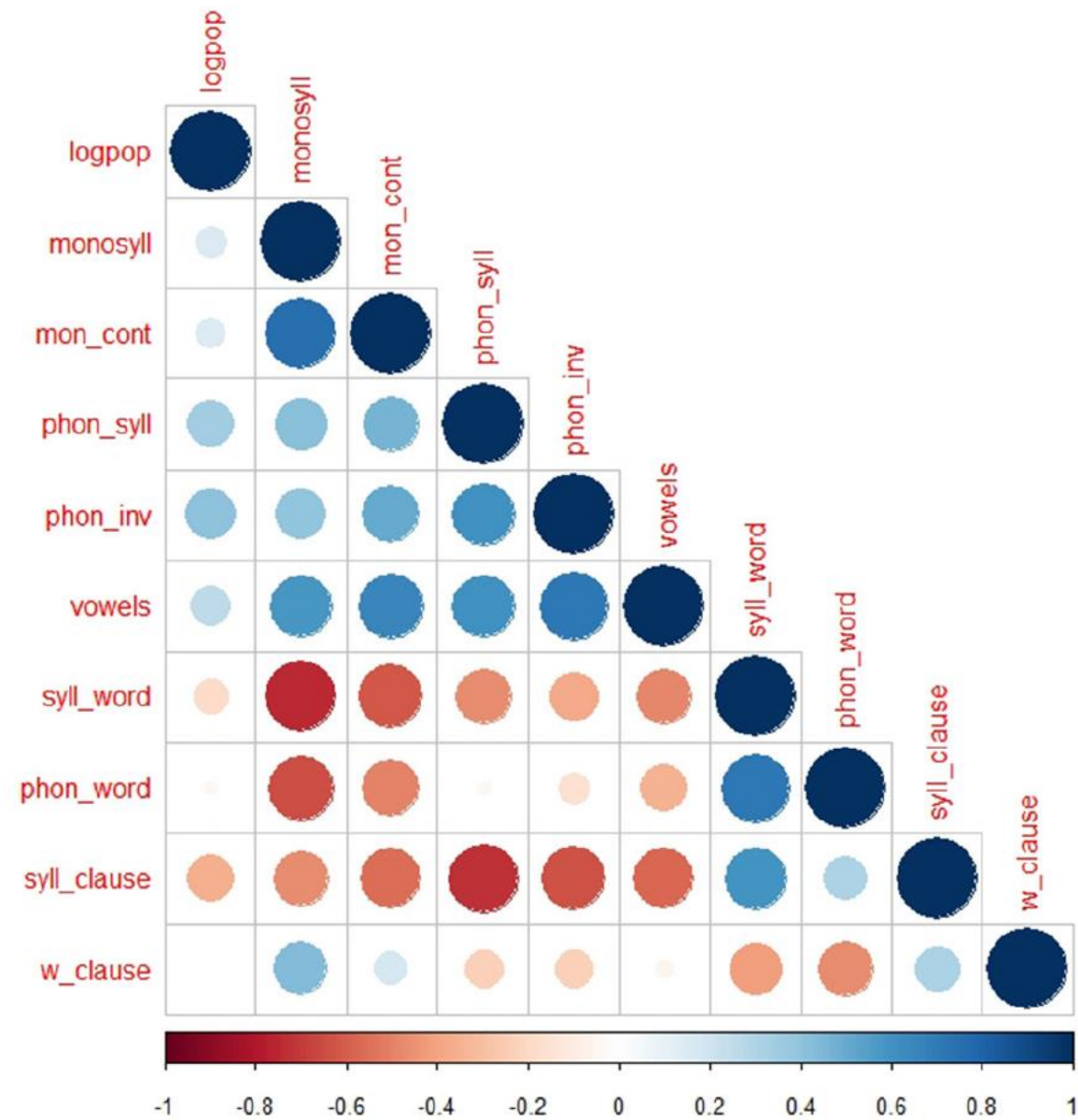


FIGURE 2 | Correlation matrix between the linguistic variables and log of population sizes.

Galton's Problem

- Sir Francis Galton (comparative anthropology): similarity between observations correlates with their **proximity in evolutionary time**
- If we don't control for this non-independence, significance can be inflated.
- Plus the problem of similarities due to language and cultural **contact**

What can we do?

- Make the languages in our sample as independent as possible
- Aggregate data by genealogical and geographic units
- Incorporate the dependencies in the random effect structure of mixed-effects regression models

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Independent languages

- Perkins (1980, 1988) and Bybee (1985) – a sample of 50 languages:
 - no more than 1 language per language phylum
 - no more than 1 language per known cultural and geographic area
- Strengths: intuitively understandable, no need for complex statistical tests
- Problems:
 - difficult to find data, we can end up testing the same set of languages
 - usually results in a relatively small sample
 - what is the right level of analysis (areas, genealogy)?

If we wanted to test a language universal relying only on our knowledge,

- How many languages could we gather data about in this room?
- How many **independent** data points would we be able to collect?

What can we do?

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Dryer's approach: genera and macroareas

	AFRICA	EURASIA	SEASIA&Oc	AUS-NEWGUI	NAMER	SAMER	TOTAL
OV&Postp	15	26	5	17	25	19	107
OV&Prep	3	3	0	1	0	0	7
VO&Postp	4	1	0	0	3	4	12
VO&Prep	16	8	15	6	20	5	70

TABLE 1. Adposition type.

Key: The numbers indicate the number of genera containing languages of the given type in the given area. The large of the two numbers for each area and for each order of verb and object is enclosed in a box. Africa includes Semitic languages of southwest Asia; Eurasia = Europe and Asia, except for southeast Asia, as defined immediately; SEAsia&Oc = Southeast Asia (Sino-Tibetan, Thai, and Mon-Khmer) and Oceania (Austronesian); Aus-NewGui = Australia and New Guinea, excluding Austronesian languages of New Guinea; NAmer = North America, including languages of Mexico, as well as Mayan and Aztecan languages in Central America; SAmer = South America, including languages in Central America except Mayan and Aztecan languages.

From Dryer 1992

Bickel's family bias method

- Assumption that all languages in a genus have the same bias is problematic.
- For every language family (at any taxonomic level), measure if it has a bias towards a feature (e.g., binomial test).
- The total proportion of families with a bias (in either direction) as against families that are diverse indicates how strong the universal pressure is.
- Example: 59 large stocks in the dataset, 39 (66%) are biased towards an A-before-P order and only 2 (3%) are biased towards the opposite. Strong evidence for the pressure of A-before-P order.

Bickel 2011

Random sampling/bootstrapping

- Levshina (2021): correlations between different corpus-driven features related to A and P
- a sample of 30 languages, mostly Indo-European
- correlation analysis:
 - 1000 random samples (one language per genus)
 - Spearman correlation coefficients are logged and then averaged
 - Permutation for null hypothesis significance testing

	Case Marking	Tight Semantics	Rigid Word Order	Verb between Subj and Obj
Case Marking		0.49 0.44	-0.67 -0.66	-0.47 -0.22
Tight Semantics	0.49 0.44		-0.16 0.26	-0.44 -0.27
Rigid Word Order	-0.67 -0.66	-0.16 0.26		0.28 0.01
Verb between Subj and Obj	-0.47 -0.22	-0.44 -0.27	0.28 0.01	

Pros and cons

- Strengths:
 - We can include as many languages as we like
- Problems:
 - What is the unit of analysis?
 - "The groups identified as genera are intended to be maximal groups with a time depth no greater than 4000 years. Because our current knowledge about the time depths of most genetic groups is rather meagre, considerable guesswork has been involved in identifying these genera." (Dryer 1992: fn2)
 - Separating areas can be tricky: "The choice of areas and where to draw their boundaries is somewhat arbitrary. (..._ The crucial question, however, is to what extent the results discussed here would have been different had a different set of areas been chosen." (Dryer 1992: fn3)
 - Family/genus size doesn't matter: Very big language families on a par with isolates
 - Low number of data points due to aggregation, loss of data

What can we do?

- Make the languages in our sample as independent as possible
- Aggregate data by genealogical and geographic units
- Incorporate the dependencies in the random effect structure of mixed-effects regression models

Mixed models

- The de facto standard in psycholinguistics and corpus-based studies of language variation
 - repeated measurements (subjects, stimuli)
 - hierarchical/nested data (sentence < text < author in a corpus)
- Mixed models combine:
 - Fixed effects: overall relationships shared across all observations
 - Linguistic variables and features we are interested in
 - Random effects: group-specific variation (e.g., participants, schools, sites, stimuli in experiments)
 - Genealogical and geographic information

Differential Object Marking (DOM)

Spanish

- Inanimate: no case markers

Ve *la* *casa*.
see.PRS.1SG DEF house
"I see the house."

- Animate and specific: case marker *a*

Ve *a* *la* *mujer*.
see.PRS.1SG OBJ DEF woman
"I see the woman."

Referential scales in DOM

Human > Animal > Inanimate

Definite > (Indefinite) Specific > Non-specific

Given > New

Pronoun > Noun



(MORE) MARKING OF P

NO (LESS) MARKING OF P

Silverstein (1976), Bossong (1991: 159), Comrie (1986: 94), Croft (2003: 132), Haspelmath (2021), Schmidtke-Bode & Levshina (2018); but Bickel et al. (2015)

Correlations between rank and marking

Table 15: Data coding for the rank-based GLM (segment)

MContinent	Family	System	RefCat	Number	Marking	Rank
Africa	Adamawa-Ubangi	Gbeya	1	SG	marked	12
Africa	Adamawa-Ubangi	Gbeya	1	NSG	marked	12
Africa	Adamawa-Ubangi	Gbeya	2	SG	marked	12
Africa	Adamawa-Ubangi	Gbeya	2	NSG	marked	12
Africa	Adamawa-Ubangi	Gbeya	3	SG	marked	3
Africa	Adamawa-Ubangi	Gbeya	3	NSG	marked	3
Africa	Adamawa-Ubangi	Gbeya	N _{high}	SG	unmarked	N _{high}
Africa	Adamawa-Ubangi	Gbeya	N _{high}	NSG	unmarked	N _{high}
Africa	Adamawa-Ubangi	Gbeya	N _{low}	SG	unmarked	N _{low}
Africa	Adamawa-Ubangi	Gbeya	N _{low}	NSG	unmarked	N _{low}

Data from AUTOTYP (Bickel et al. 2023)
<https://doi.org/10.5281/zenodo.7976754>

Schmidtke-Bode & Levshina 2018

Multilevel logistic model

- Response variable: marked or unmarked P (object)
- Fixed effects:
 - Rank on the referential hierarchy
 - 1st or 2nd person > 3P pronouns > High-ranking noun (animate, definite, given...)
>Low-ranking noun (inanimate, indefinite, new...)
 - Number
 - Macrocontinent
 - and interactions between those
- Random intercepts: "System" (language) and Family

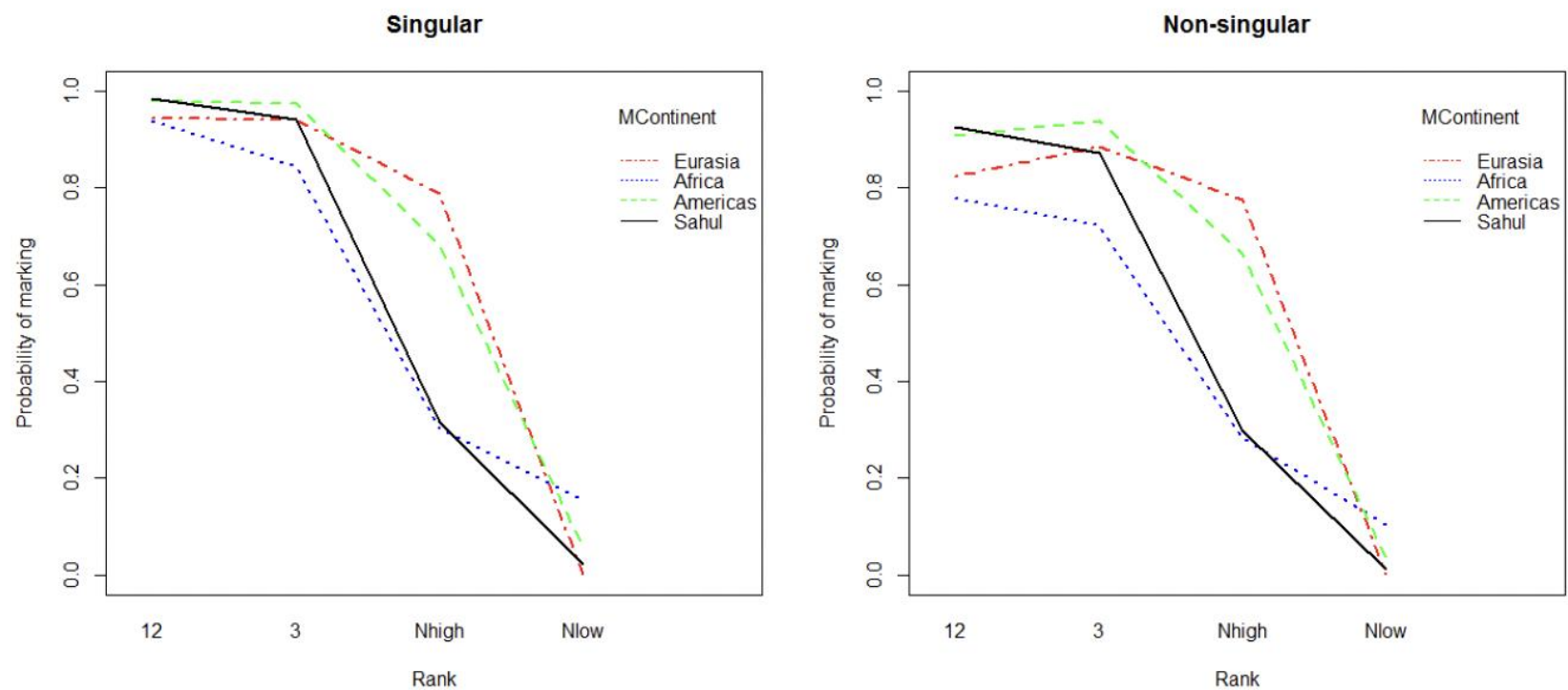
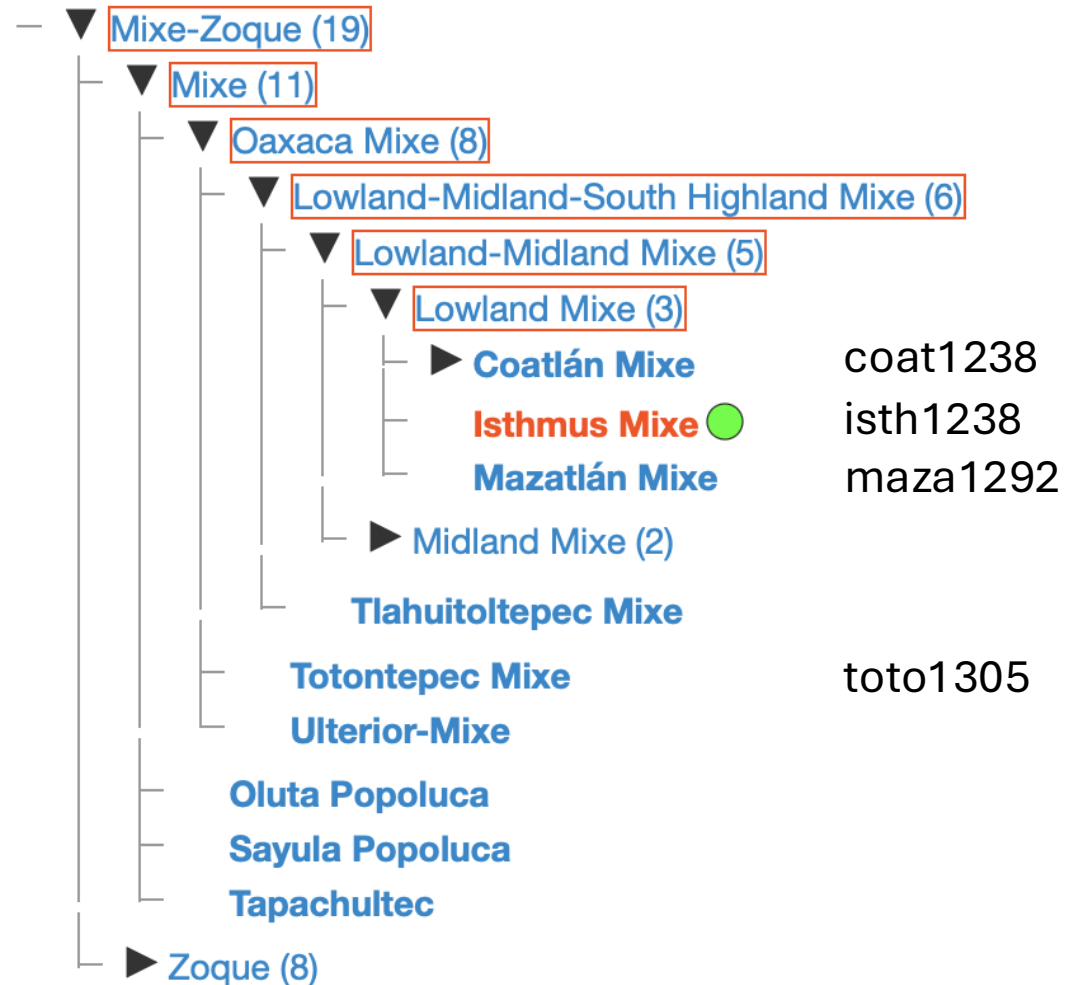


Figure 1: Influence of ranks on the probability of marked Ps in the singular (left) and the non-singular (left), by macro continent.

Pros and cons of using groups as random effects

- Strengths:
 - We can use all languages as data points, no loss of data
 - A well-established statistical method
 - Flexibility: all kinds of variables
- Problems:
 - choosing the levels for groupings
 - loss of information from phylogenetic trees

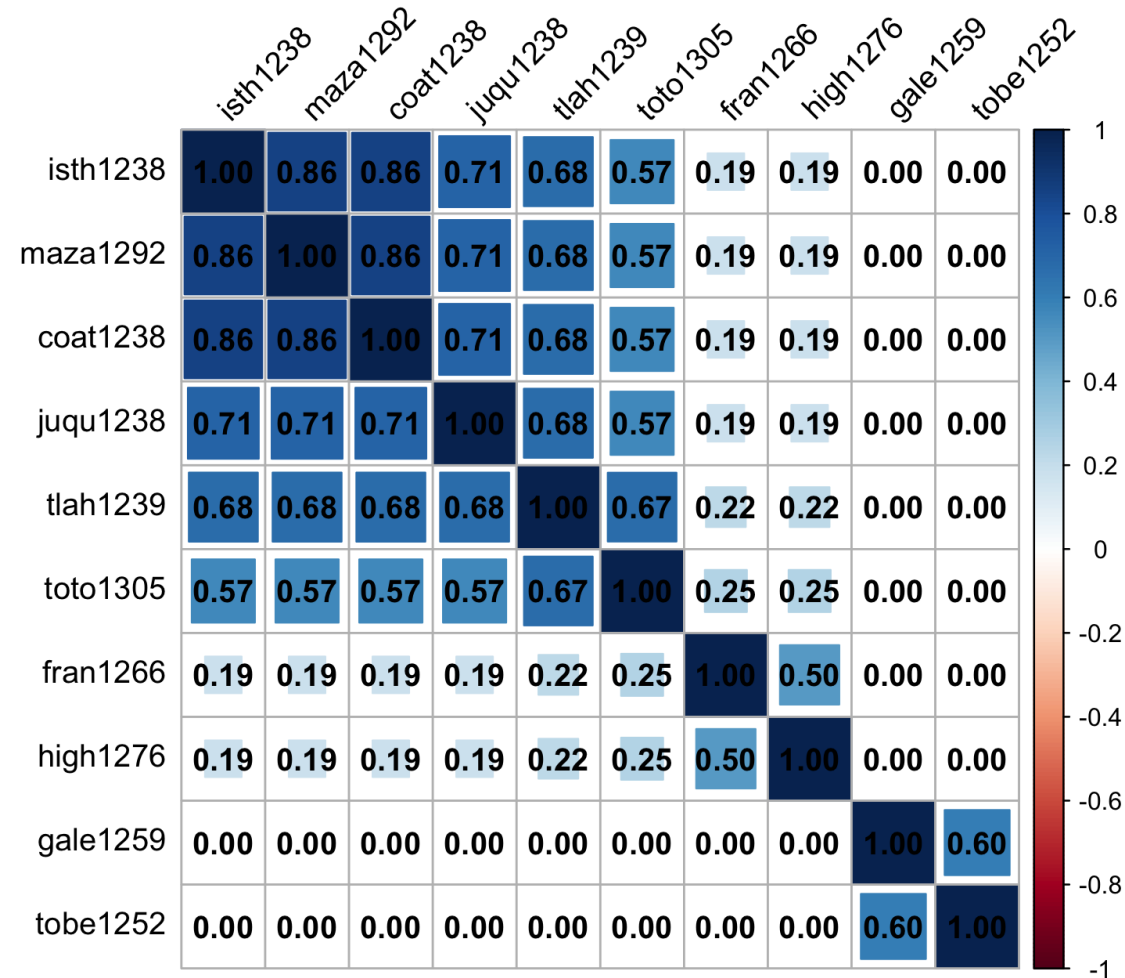
Example: Mixe-Zoque languages



Hierarchical phylogenetic regression

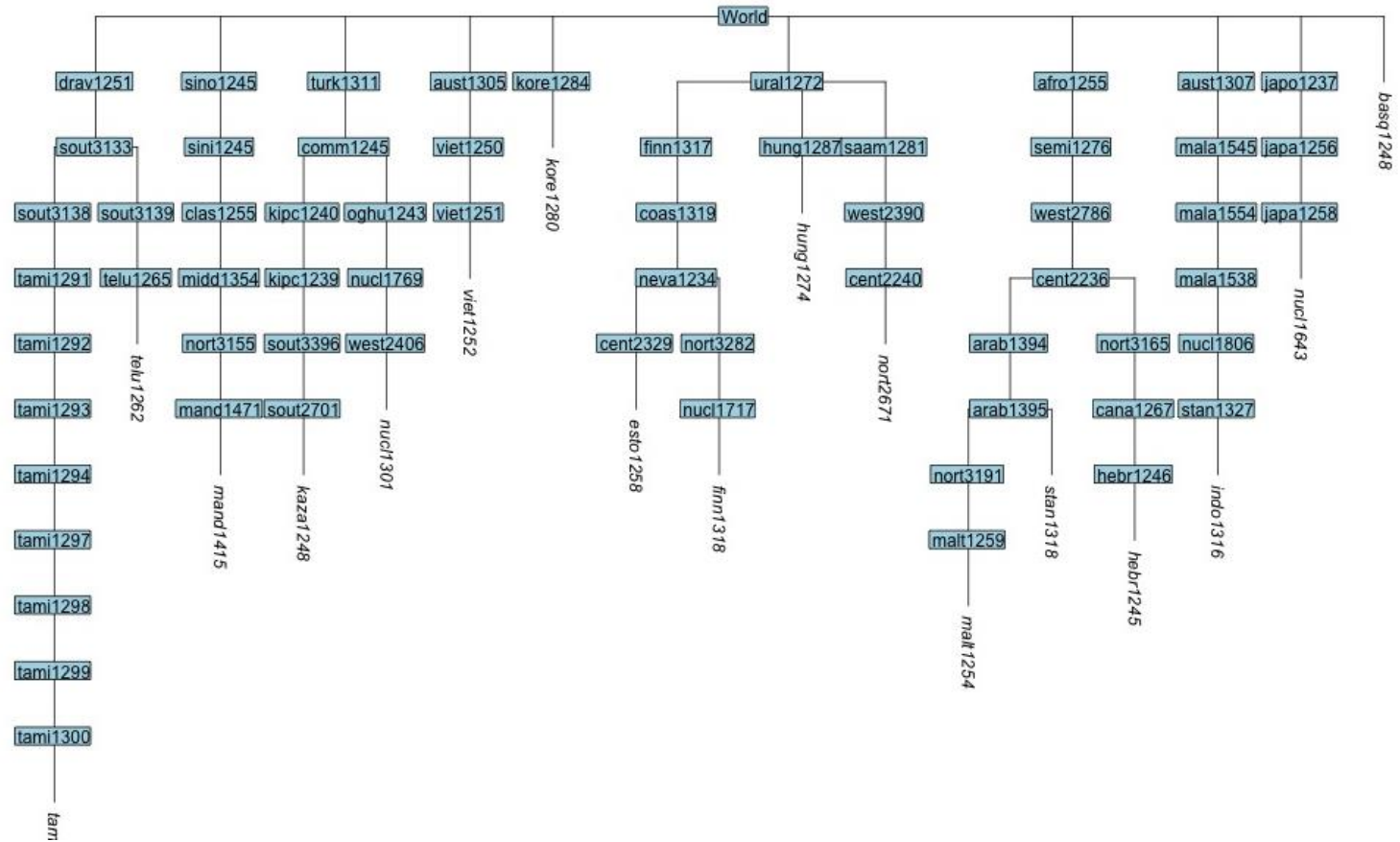
- Originally developed in evolutionary biology (*Hadfield & Nakagawa 2010*)
- Convenient implementation in R package *brms* (*Bürkner 2024*)
- Random structure reflects correlations between species (langoids belonging to different genera and families).

Phylolinguistic information from Glottolog



One "supertree"?

- A problem: we do not know the depth at which all language families merge.
- Fortunately, we don't need to operate on a tree. We can assign zero correlations in the covariance matrix if two languages belong to different language families.



What about language contact?

- Languages that are close geographically are more likely to be in contact.
- We can use geographic coordinates to control for spatial autocorrelations
- A solution is to create a variance-covariance matrix, such that languages that are closer influence each other more.
- The covariances (correlations) can be calibrated: is diffusion likely only locally (e.g., hundreds of kilometers), or also across larger regions?

A problem with distances

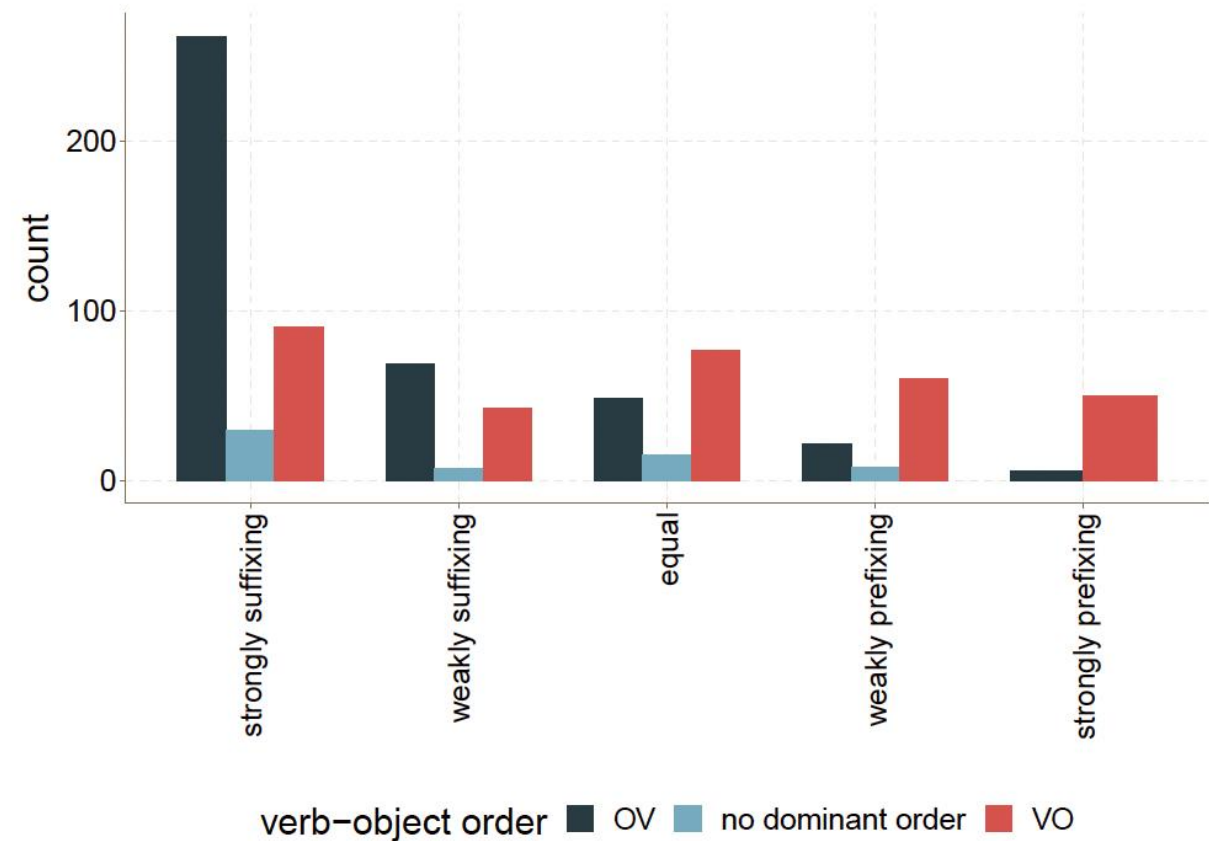
- Problem (Dryer 2018): if two languages are 100 km apart, how likely are they to display common features due to contact?
 - If in Siberia → more likely
 - If in New Guinea → less likely
- A solution is to use Gaussian Processes (GP) based on latitude and longitude (Guzmán Naranjo & Becker 2022)
 - two close languages will have a strong influence over each other
 - two observations far apart will have less influence over each other
 - the effect is inferred automatically from the data → the effect of distance can vary across areas

Pros and cons of Bayesian phylogeographic regression

- Strengths:
 - many data points
 - flexibility
 - fine-grained representations of geological and geographic information
- Problems:
 - often computationally intensive (we usually cannot test random slopes)
 - uncertainty in phylogenies (what is the best tree?). Possible solution: apply regression on a set of possible trees, incorporate the uncertainty in the regression parameters (not part of this course)
 - actual interaction between linguistic communities depends on more factors than just geographic distance between them based on 2-dimensional coordinates; coordinates are not very informative for languages spoken in a large area

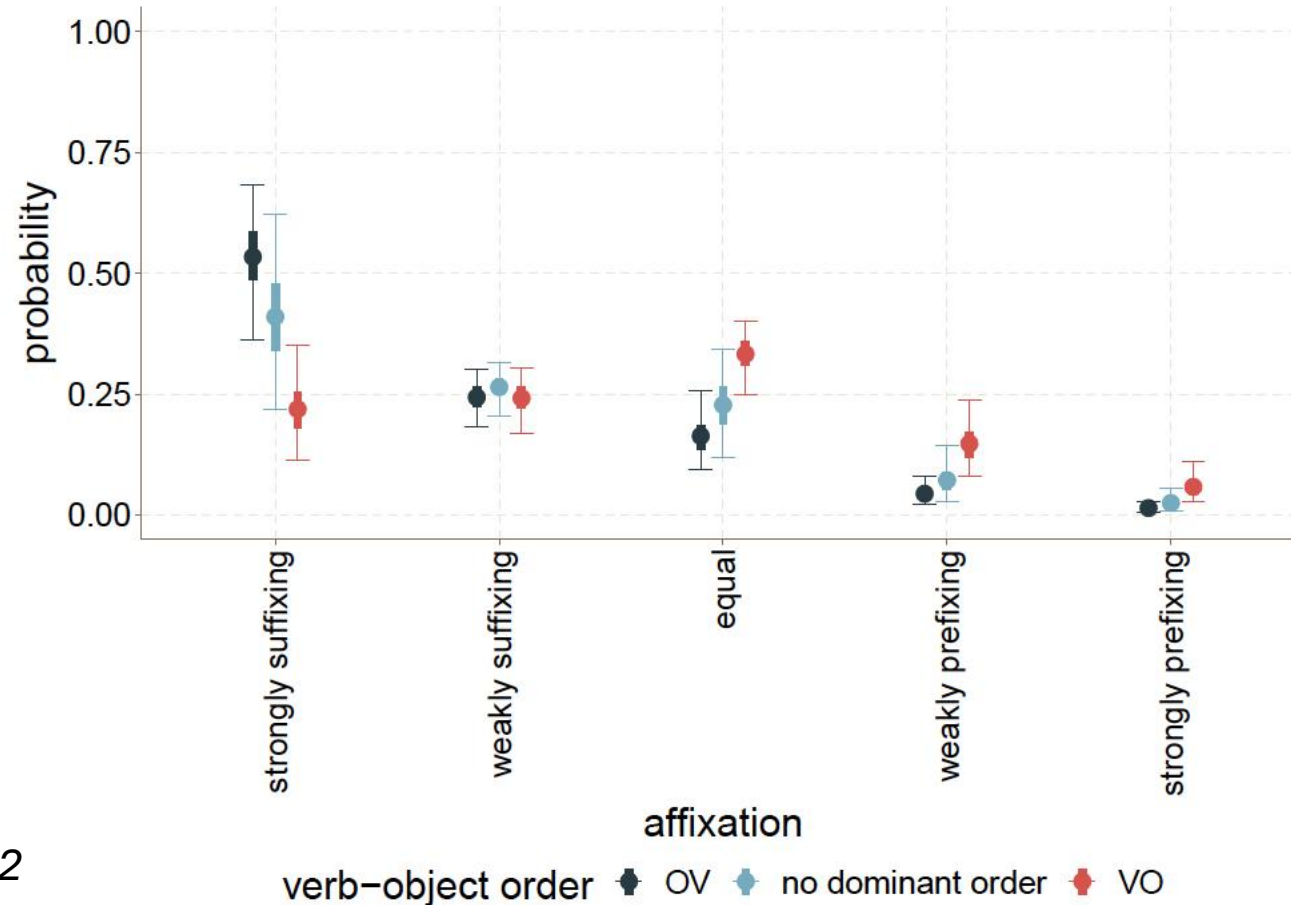
Comparison of methods

- Guzmán Naranjo & Becker (2022): a study of correlation between affix position and Object-Verb order
- Hypothesis: OV languages strongly prefer suffixation, whereas VO languages can have both.



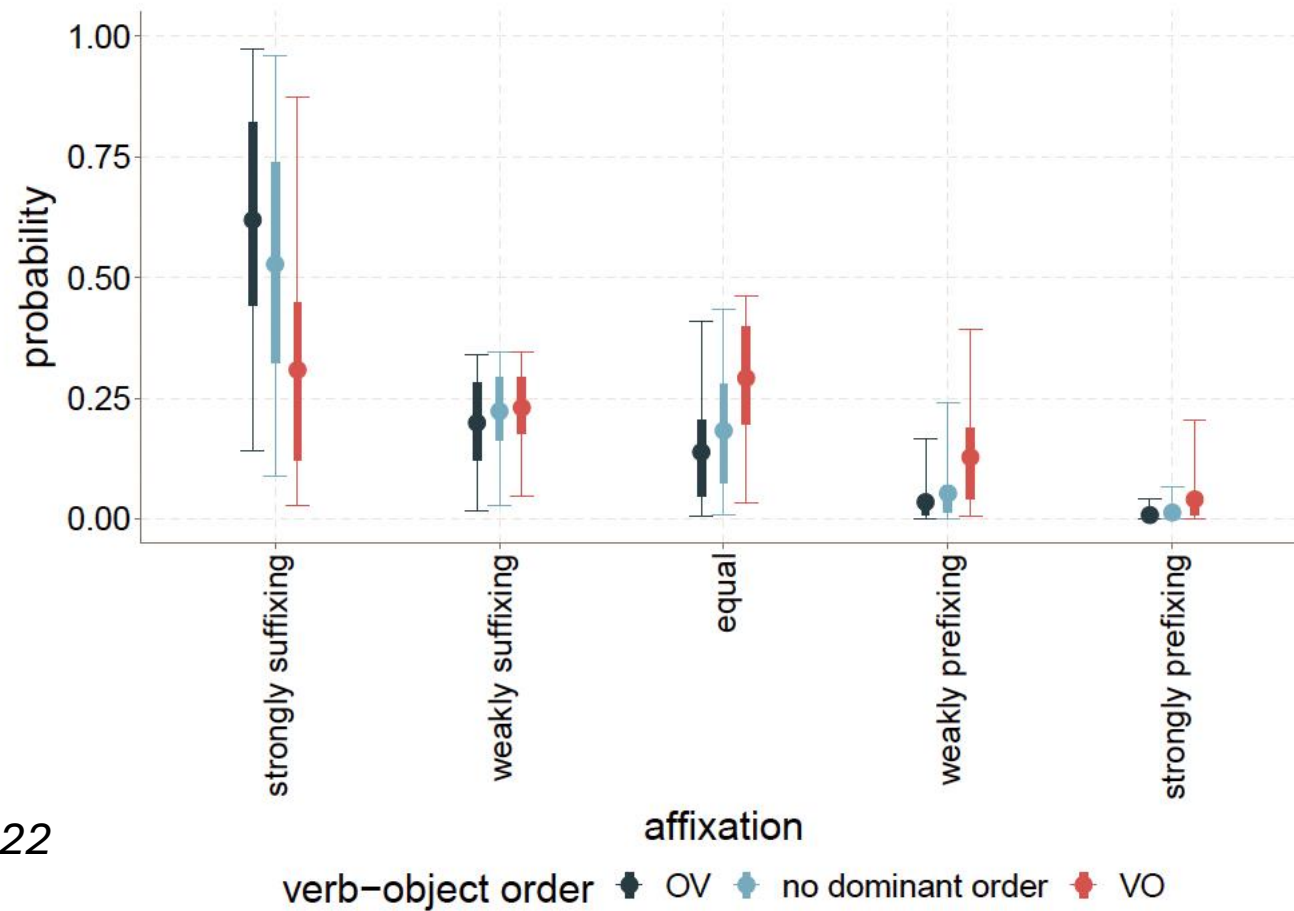
Model with "normal" random intercepts

affixation ~ VO-order + (1|family) + (1|macroarea)



Phylogeographic model

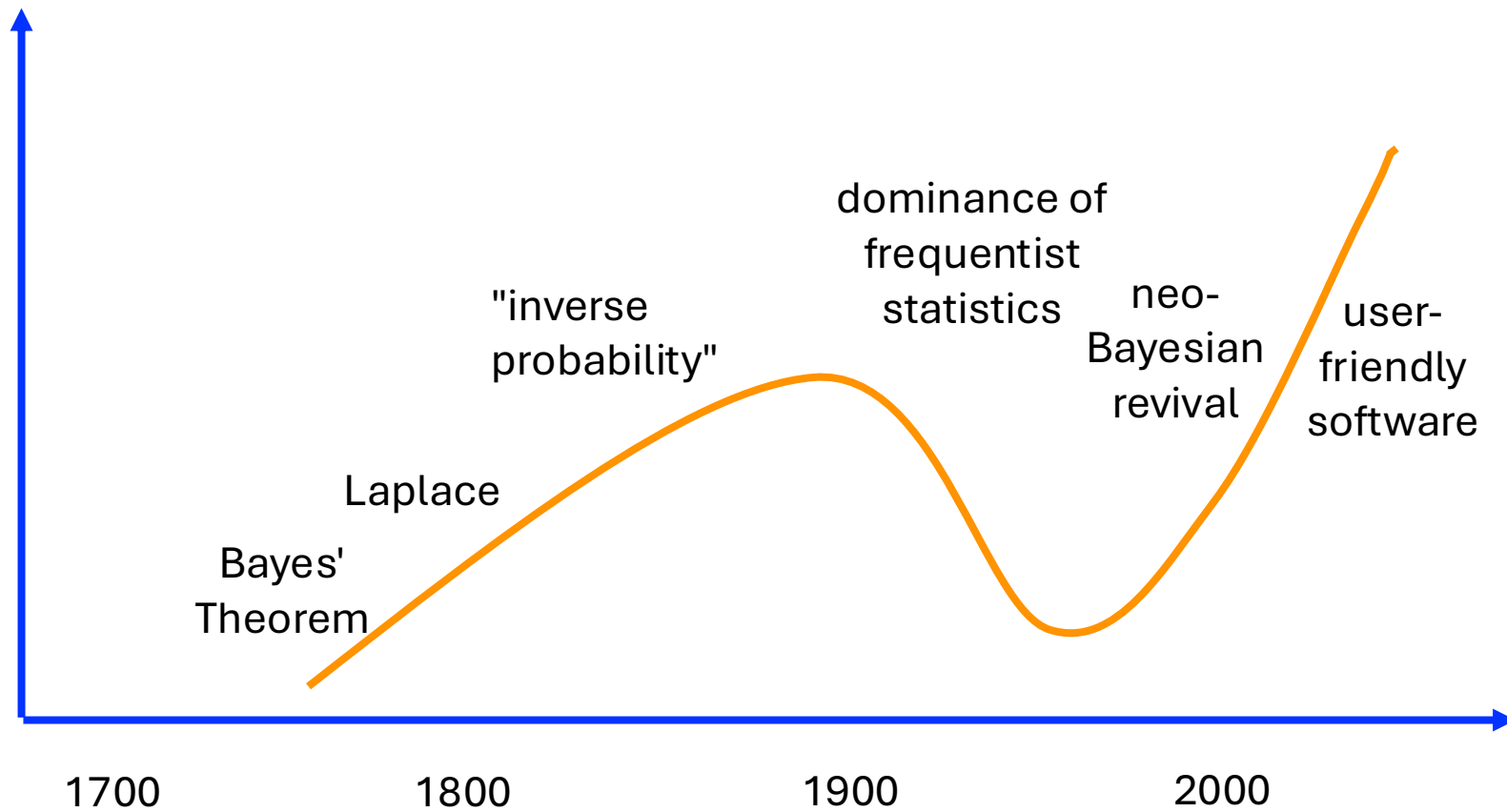
affixation ~ VO-order + micro-family covariance matrix + Gaussian Processes by macroarea



Outline

- Cross-linguistic generalizations and problems of testing them
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Bayesian statistics timeline



What are probabilities?

Frequentist statistics

Probabilities as long-run frequencies

If a coin is fair, and we toss the coin very many times, we expect to get the heads 50% of the times and the tails 50% of the times. The more times we repeat the experiment, the closer the observed proportions of heads and tails will be to 50%.

Bayesian statistics

Probabilities as beliefs

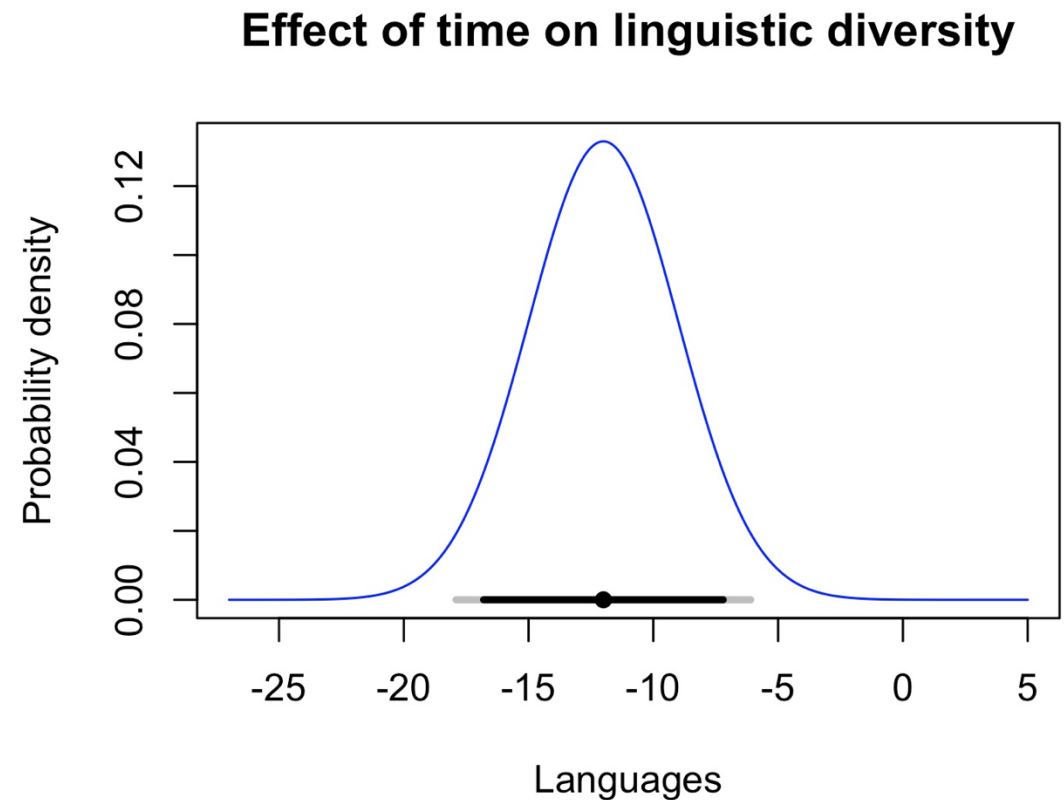
Beliefs - an umbrella term for expectations, opinions, assumptions, convictions... For every value of a parameter, one can think of its probability.

What are betas in regression?

Frequentist statistics	Bayesian statistics
Parameters as unknown but fixed constants A coefficient estimate in regression is one number.	Parameters as random variables represented as probability distributions A coefficient estimate is a probability distribution.

Illustration: loss of linguistic diversity

- Bromham et al. (2022): at least one language is lost per month if we do not intervene.
- An imaginary regression coefficient of Time (in years):



Basic terms

$$P(\theta | \text{data}) = P(\text{data} | \theta) * P(\theta) / P(\text{data})$$

- $P(\theta | \text{data})$ - the posterior probability of θ given the knowledge of the data. In other words, a posterior distribution is the degree of belief in every value of the parameter after considering the data.
- $P(\text{data} | \theta)$ - the probability of the data given θ , also known as the likelihood of θ for given data (*Fisher 1922*).
- $P(\theta)$ - the prior distribution, or the distribution of prior probabilities of θ without the knowledge of the data; in other words, the degree of belief in every specific value of the parameter before considering the data.
- $P(\text{data})$ - evidence, or marginal likelihood. It is usually a normalizing constant that is necessary to make sure that the posterior distribution integrates to one.

Posterior distribution $\propto$ Likelihood $\times$ Prior distribution

Example

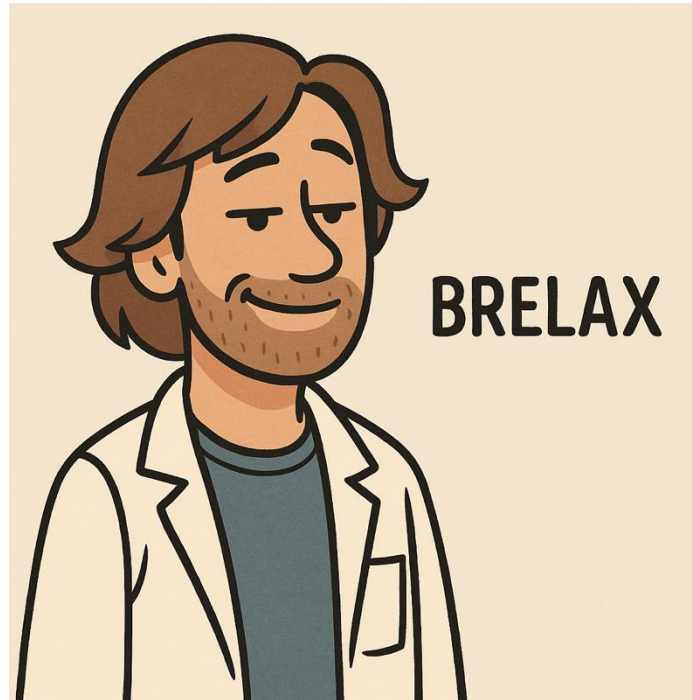
- Effect of word surprisal in bits on regression-path duration

Researchers

Assertia



Brelax



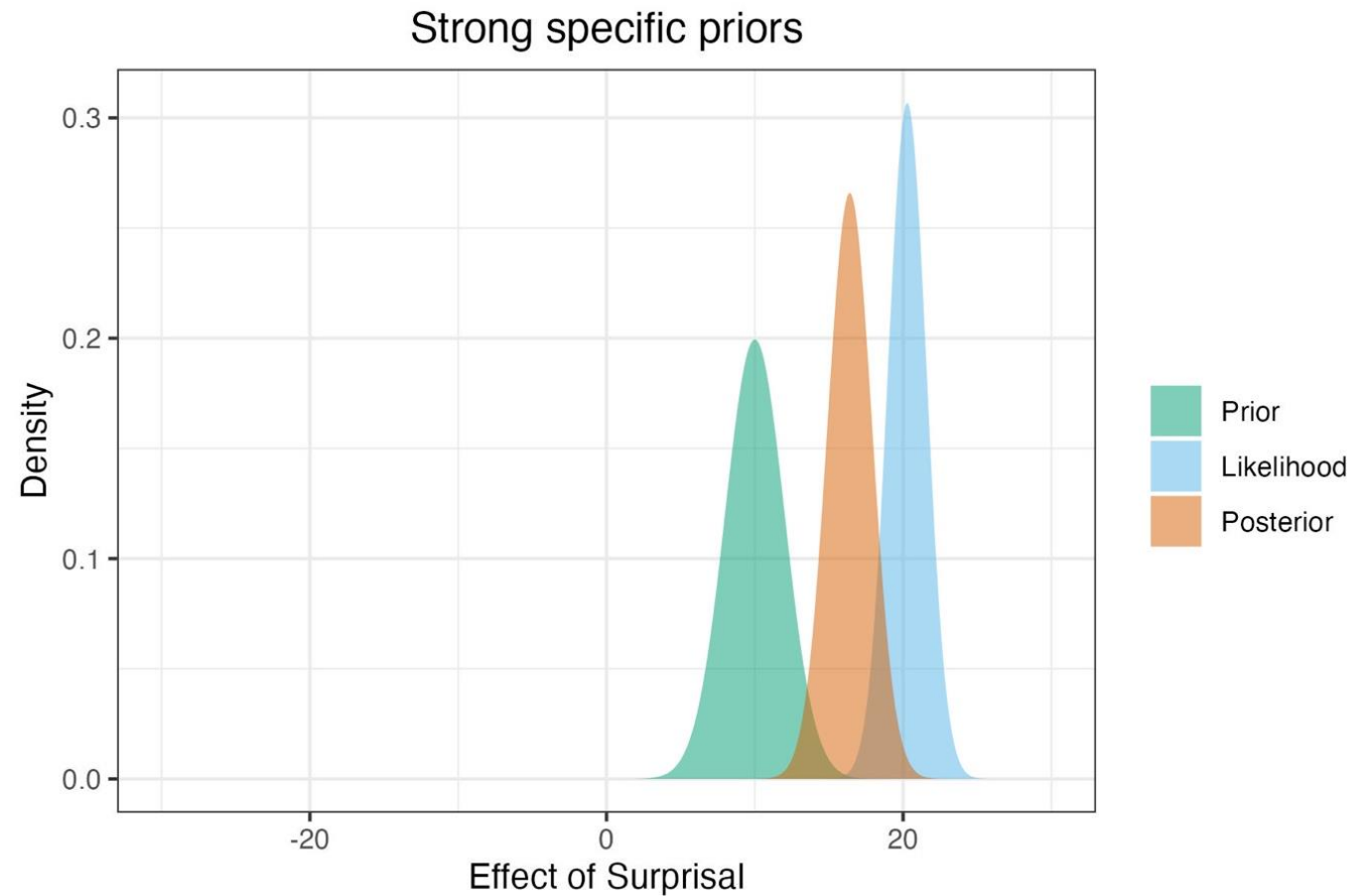
Causy



Informative priors

- Assertia has strong beliefs about the effect of surprisal: one bit of surprisal adds 10 ms of duration
- She uses informative, strong specific priors.

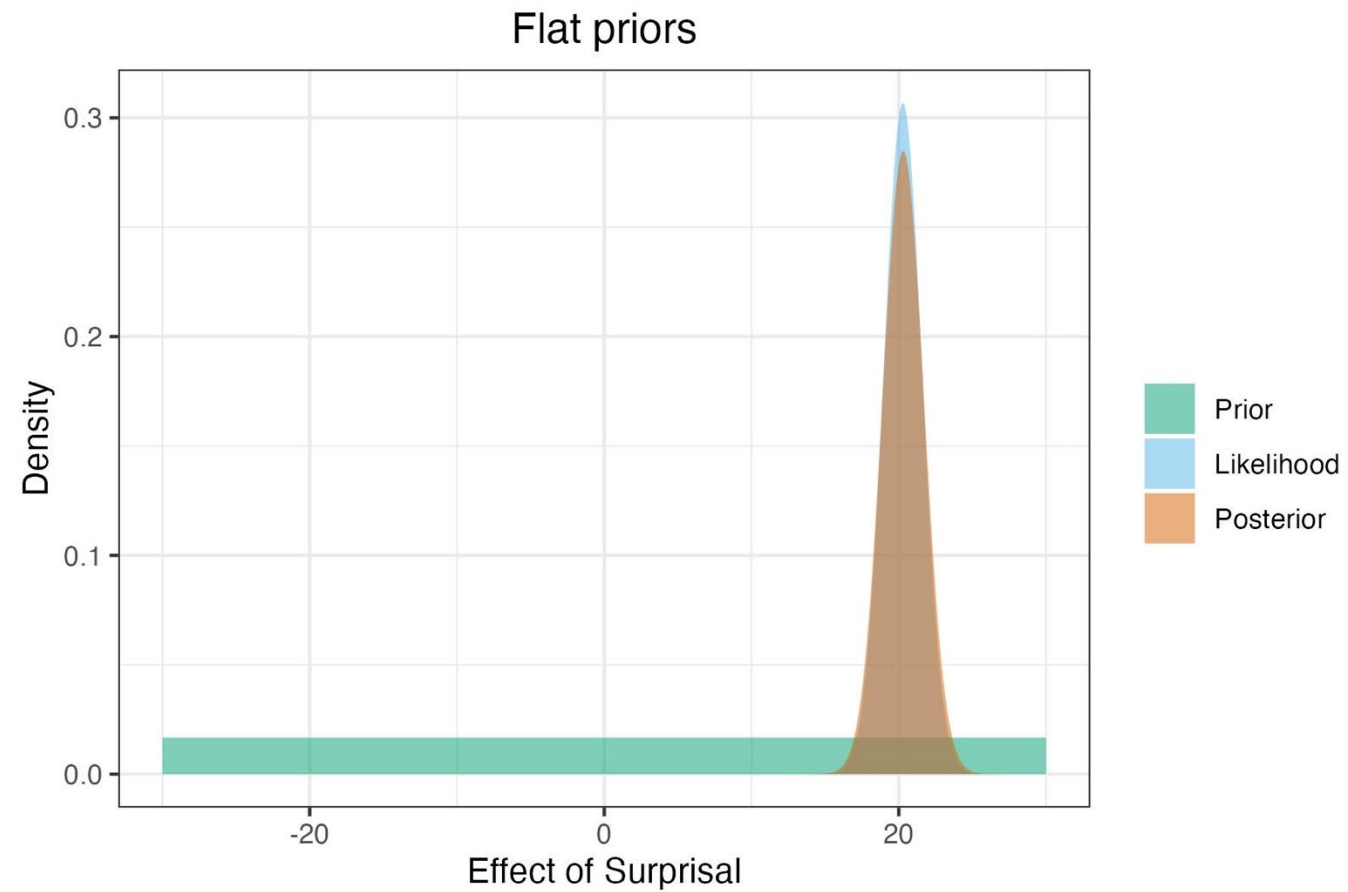
Assertia's results



Brelax

- Brelax: all values are equally likely.
- He uses uniform (vague, diffuse, flat) priors

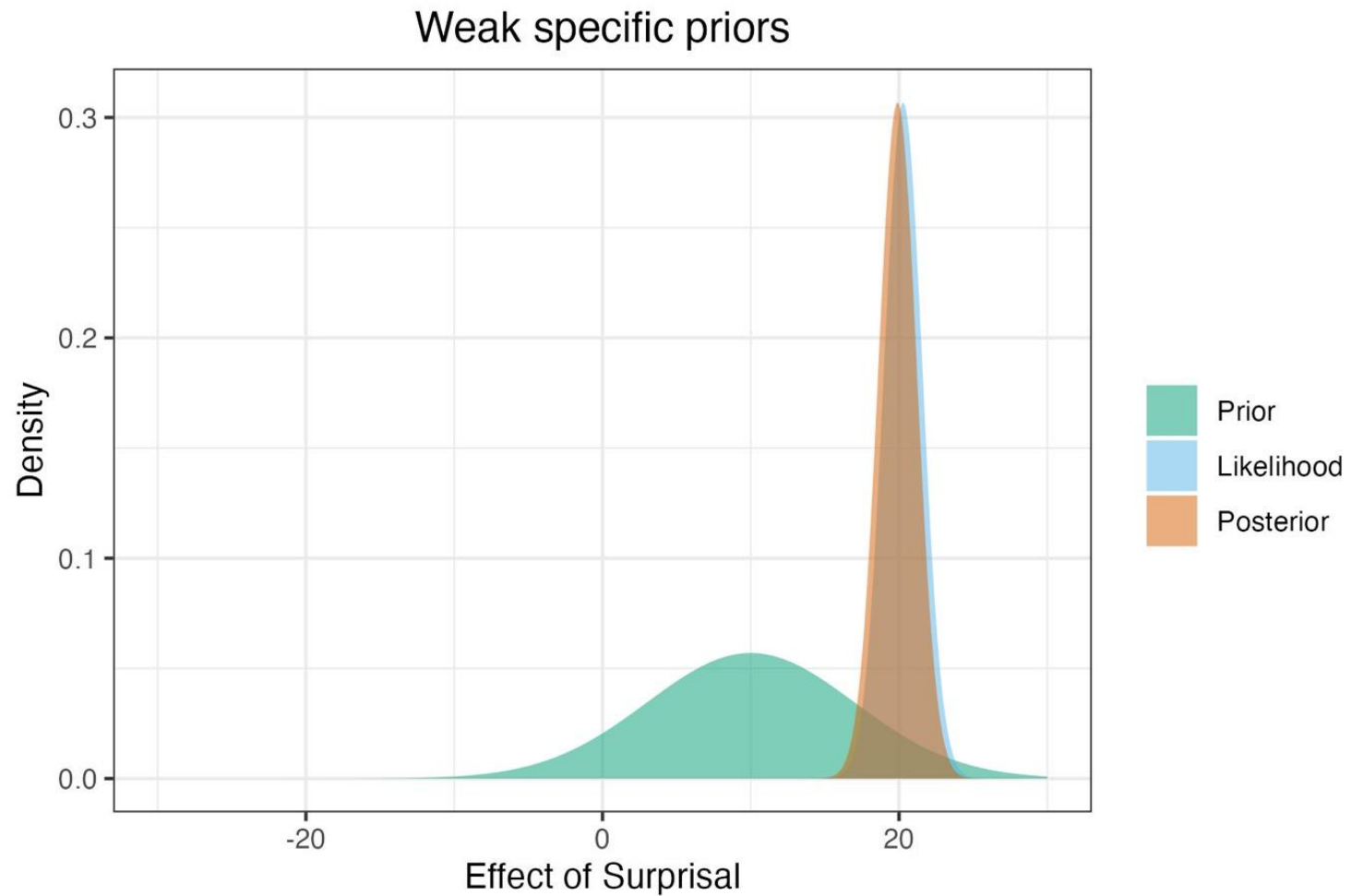
Brelax' results



Causy

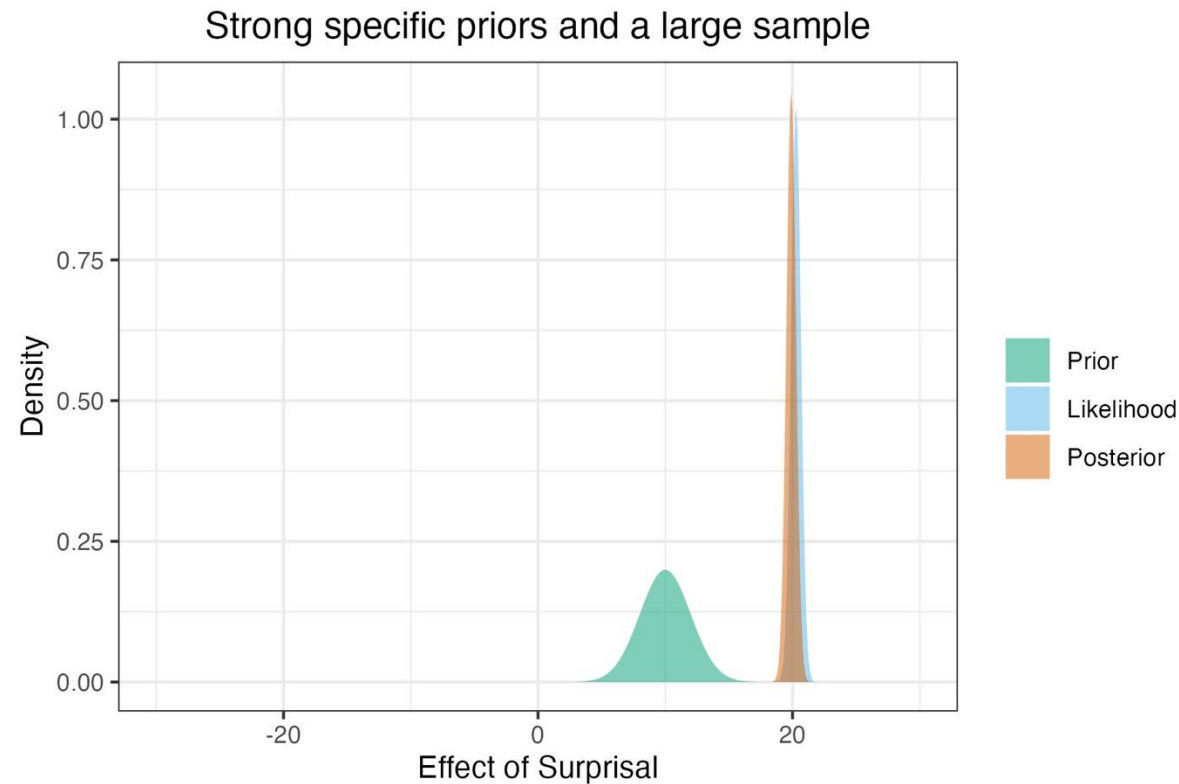
- Causy has specific priors around 10 ms, like Assertia, but these priors allow for more variation because the distribution is more spread out.
- Such priors are called weakly informative priors.

Causy's results



More data

- If Assertia gets some extra data and uses the same priors, this is what she obtains:



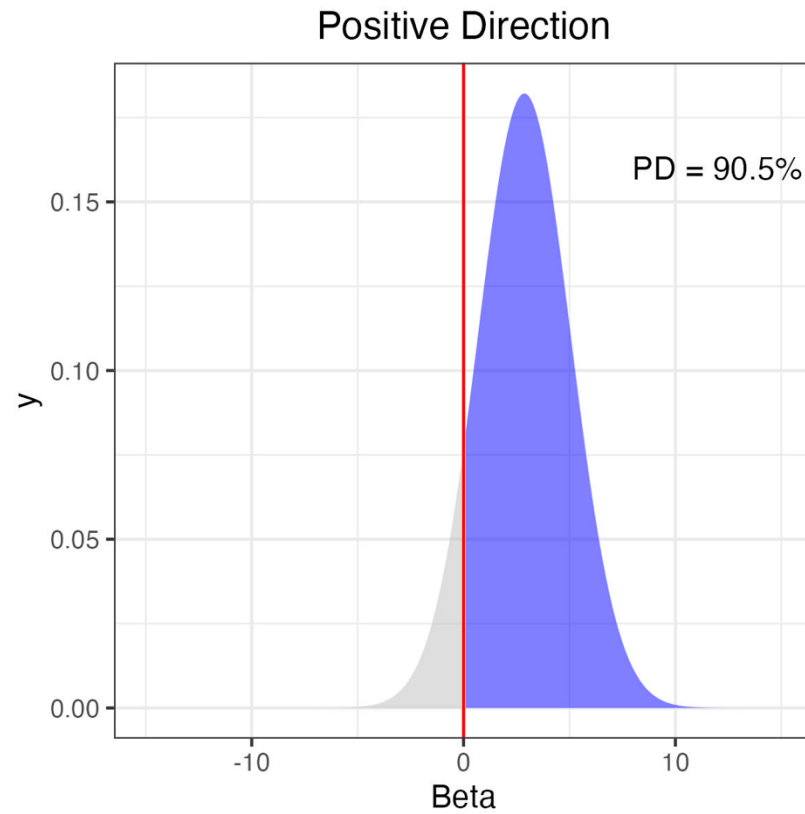
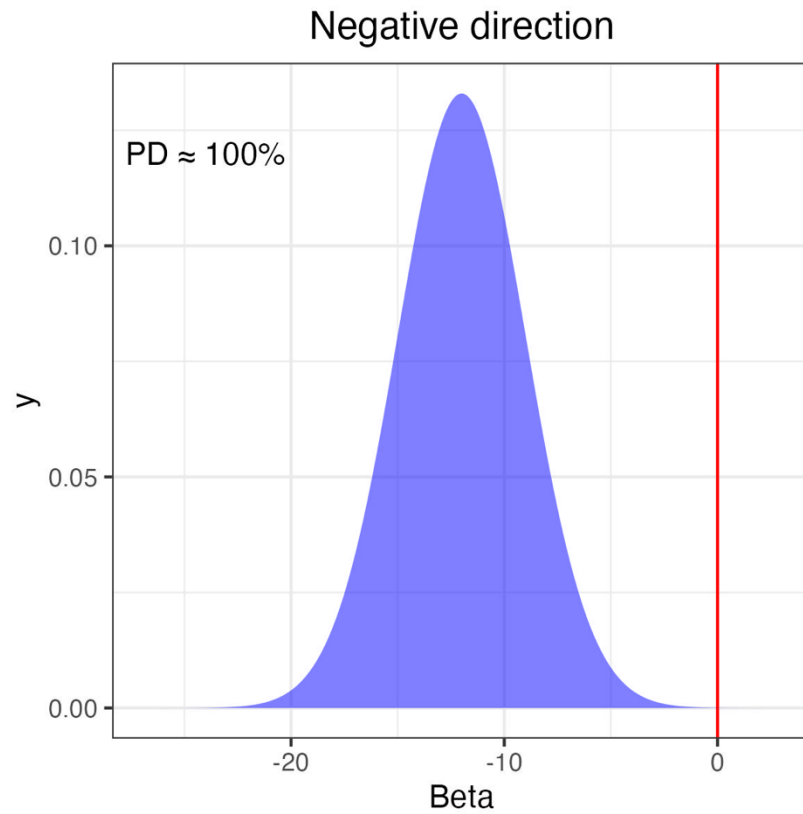
MCMC: a key ingredient

- Monte Carlo Markov Chains estimation enables us to estimate posterior probability distributions, which would not be possible otherwise in most practical cases.
- Markov Chains are models in which the probability of an event depends only on the state attained as the result of the previous event.
- Monte Carlo is a family of methods related to random sampling from a distribution.
- We can sample directly from the posterior without assuming any shape (beta, Gaussian, etc.)
- This can be done with Stan, a platform for high-performance statistical computation: <https://mc-stan.org/> , named after Stanislaw Ulam, one of the inventors of MCMC.

Hypothesis testing

Frequentist statistics	Bayesian statistics
Null hypothesis significance testing: Can we reject the null hypothesis? Tools: p-values, confidence intervals.	Testing the research hypothesis directly: Can we support the research hypothesis? Tools: ROPE, Probability of Direction Which hypothesis is more supported by the data? Tools: Bayes Factor, ROPE...

Probability of Direction



Intervals

Frequentist statistics

Confidence intervals:

If we calculate a 95% confidence interval in this way many times on repeated samples, the true mean will be between the boundaries 95% of the time.

Bayesian statistics

Credible intervals:

the parameter of interest (e.g. a regression coefficient) falls in this interval with the probability of ... %

Different types: a) Highest Posterior Density
b) Equal-Tailed Interval

Model selection and comparison

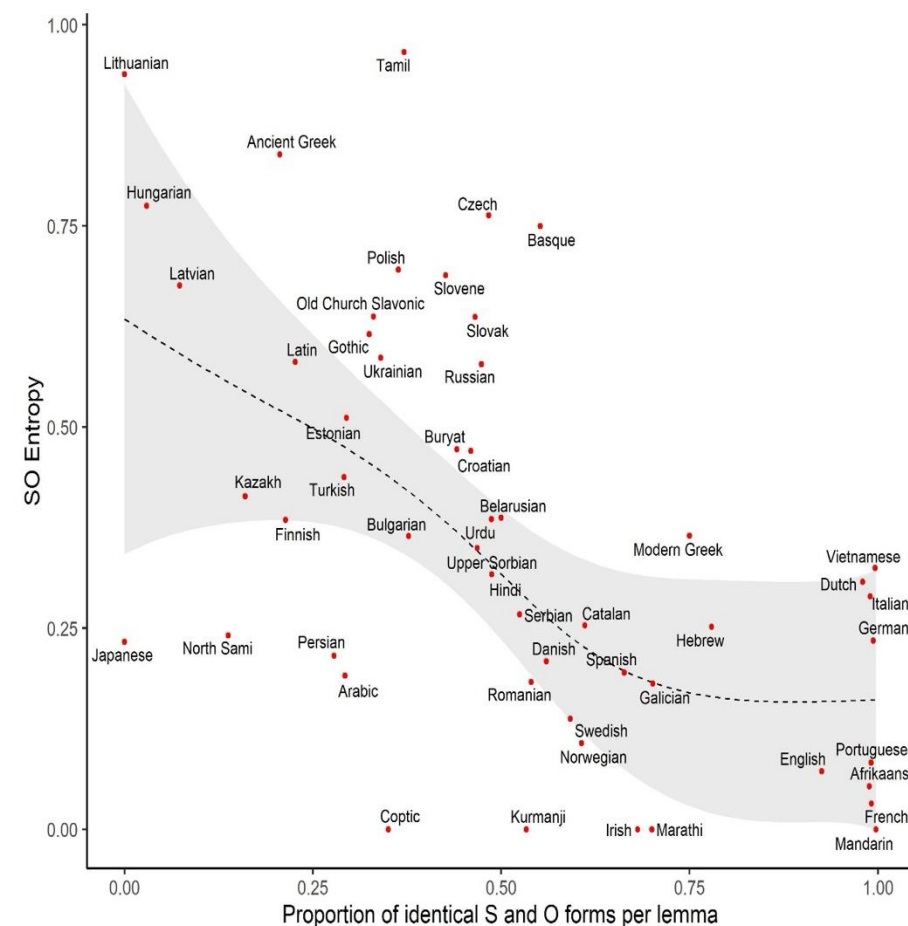
Frequentist statistics	Bayesian statistics
Likelihood ratio tests on nested models; information criteria (AIC, BIC)	comparison of ELPD (Expected Log pointwise Predictive Density), using leave-One-Out (LOO) cross-validation, Bayesian information criteria (e.g., WAIC)

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Case study

- Languages without case marking tend to have rigid word order, whereas languages with case marking allow for more flexibility.
- Levshina (2019, LingTyp): focus on S and O in Universal Dependency corpora.
- A negative correlation between the proportion of nominal lemmas that have the same S and O forms in the corpus, and entropy of SO order.



Entropy

- Proportions of nsubj + obj and obj + nsubj (only common nouns) in a transitive clause
- The higher entropy H , the greater the variability

$$H(X) = -\sum_{i=1}^2 P(x_i) \log_2 P(x_i)$$

